

Detailed solutions for the uploaded test:  PlacementAssessmentTest1-2

1

(a)

$$\frac{4}{5} \left( 10 - \frac{5}{4} \right) = \frac{4}{5} \left( \frac{40}{4} - \frac{5}{4} \right) = \frac{4}{5} \cdot \frac{35}{4} = 7$$

(b)

$$\left( 4 + \frac{1}{3} \right) \left( 1 - \frac{3}{4} \right) = \frac{13}{3} \cdot \frac{1}{4} = \frac{13}{12}$$

2

$$64^{1/3} = 4$$

$$(-27)^{1/3} = -3$$

$$-\left( \frac{1}{27} \right)^{1/3} = -\frac{1}{3}$$

3

$$\begin{aligned} & (6x - 3)(x^2 - x + 1) \\ &= 6x^3 - 6x^2 + 6x - 3x^2 + 3x - 3 \\ &= \boxed{6x^3 - 9x^2 + 9x - 3} \end{aligned}$$

4

$$\frac{x - 9}{x^2 - 81}$$

Factor the denominator:

$$x^2 - 81 = (x - 9)(x + 9)$$

$$\frac{x-9}{(x-9)(x+9)} = \boxed{\frac{1}{x+9}}$$

Restrictions:  $x \neq 9, -9$ .

5

$$|7x| = 4$$

$$7x = 4 \quad \text{or} \quad 7x = -4$$

$$x = \frac{4}{7}, -\frac{4}{7}$$

$$\boxed{-\frac{4}{7}, \frac{4}{7}}$$

6

$$|3x + 5| < 7$$

$$-7 < 3x + 5 < 7$$

Subtract 5:

$$-12 < 3x < 2$$

Divide by 3:

$$-4 < x < \frac{2}{3}$$

$$\boxed{\left(-4, \frac{2}{3}\right)}$$

7

$$-4x + y = 2$$

Solve for  $y$ :

$$y = 4x + 2$$

So the slope is:

$$\boxed{4}$$

The  $y$ -intercept is:

$$(0, 2)$$

8

$$f(x) = x^3 + 2x$$

$$f(-2) = (-2)^3 + 2(-2) = -8 - 4 = -12$$

$$f(-1) = (-1)^3 + 2(-1) = -1 - 2 = -3$$

$$f(0) = 0^3 + 2(0) = 0$$

$$f\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^3 + 2\left(\frac{1}{2}\right) = \frac{1}{8} + 1 = \frac{9}{8}$$

Answers:

$$-12, -3, 0, \frac{9}{8}$$

9

$$x = y^8$$

For  $x > 0$ , there are two possible  $y$ -values:

$$y = \sqrt[8]{x} \quad \text{and} \quad y = -\sqrt[8]{x}$$

So  $y$  is not uniquely determined by  $x$ .

is not a function

10

$$f(x) = \sqrt{x - 6}$$

For the square root to be real:

$$x - 6 \geq 0$$

$$x \geq 6$$

Domain:

$$[6, \infty)$$

Since square roots are never negative:

$$\boxed{[0, \infty)}$$

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$$f(x) = x^2 - 4x + 5$$

Complete the square:

$$x^2 - 4x + 5 = (x^2 - 4x + 4) + 1$$

$$f(x) = (x - 2)^2 + 1$$

Vertex form:

$$\boxed{f(x) = (x - 2)^2 + 1}$$

Vertex:

$$\boxed{(2, 1)}$$

x-intercepts:

$$(x - 2)^2 + 1 = 0$$

$$(x - 2)^2 = -1$$

No real solution, so:

$$\boxed{\text{DNE}}$$

y-intercept:

$$f(0) = 0^2 - 4(0) + 5 = 5$$

$$\boxed{(0, 5)}$$

Domain:

$$\boxed{(-\infty, \infty)}$$

Range:

$$\boxed{[1, \infty)}$$

12

$$f(x) = 2^x$$

This is increasing and passes through  $(0, 1)$ .

$$g(x) = 2^{-x}$$

This is decreasing and also passes through  $(0, 1)$ .

The correct graph is the one where both curves stay positive, meet at  $(0, 1)$ ,  $f$  increases, and  $g$  decreases. That is the bottom-right graph.

13

$$t = \frac{3\pi}{2}$$

On the unit circle, this angle points straight down.

$$P(x, y) = (0, -1)$$

14

$$\frac{7\pi}{6}$$

Convert radians to degrees:

$$\begin{aligned} \frac{7\pi}{6} &\cdot \frac{180^\circ}{\pi} \\ &= 7 \cdot 30^\circ \\ &= \boxed{210^\circ} \end{aligned}$$

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Polar coordinate:

$$\left(-4, \frac{4\pi}{3}\right)$$

A negative radius means move in the opposite direction, so add  $\pi$  to the angle:

$$\begin{aligned} \frac{4\pi}{3} + \pi &= \frac{7\pi}{3} \\ \frac{7\pi}{3} &= 2\pi + \frac{\pi}{3} \end{aligned}$$

So the point is equivalent to:

$$\left(4, \frac{\pi}{3}\right)$$

This is the point in Quadrant I at radius 4, angle  $\frac{\pi}{3}$ . In Cartesian form:

$$x = 4 \cos \frac{\pi}{3} = 2$$

$$y = 4 \sin \frac{\pi}{3} = 2\sqrt{3}$$

$$\boxed{(2, 2\sqrt{3})}$$

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From the graph:

$$P = (2, 4), \quad Q = (7, 3)$$

Vector component form:

$$\vec{v} = \langle 7 - 2, 3 - 4 \rangle$$

$$\boxed{\langle 5, -1 \rangle}$$

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Sequence:

$$19, 12, 5, -4$$

Check differences:

$$12 - 19 = -7$$

$$5 - 12 = -7$$

$$-4 - 5 = -9$$

The differences are not constant, so the sequence is not arithmetic.

$\boxed{\text{No, the sequence is not arithmetic.}}$

$$\boxed{d = \text{DNE}}$$